

Soft Spaces - Phase 2

Scientific status after v25.6 and rationale for moving to v25.7

Purpose of this note. This is a research checkpoint intended to preserve the reasoning developed during the 9-qubit Phase 2 experiments. It records what v25.6 established, what it did not establish, how the physical diagnostics should be interpreted, and why the next step should be v25.7 rather than simply extending the local scan.

1. The change in the project

Phase 1 was primarily empirical: candidate structures were found by scanning Hilbert space and comparing REAL with NULL controls. This was useful for discovering reproducible structure, but it did not by itself explain *why* a particular subspace should be special.

The work from v25.4 through v25.6 changes the question. Instead of asking only, 'Where do the numbers look unusual?', we are beginning to ask, 'Which physical properties make a subspace resistant to perturbation, and can those properties be expressed mathematically?'

This is the central transition from a largely empirical hotspot search toward a physically motivated Soft-Space formalism.

2. Why v25.4 was not enough

The continuation ranker v25.4/v25.4.1 attempted to predict 9-qubit successors of the 8-qubit anchor around 157-158. Dimension normalization corrected an important problem, but the ranking still showed that the original coordinate 157 was not uniquely selected by the static continuation score. With 3 x 25 seeds, coordinate 157 ranked 96/255, while the best coordinate within +/-8 of the old anchor was 165 at global rank 51.

That was not evidence that the earlier result was false. It was evidence that the continuation expression was incomplete: static similarity and empirical recurrence alone were not sufficient to identify the relevant physics.

3. The physical diagnostics introduced in v25.5-v25.6

K_delta - measures how strongly a perturbation couples the chosen subspace P to the outside space Q. The favourable direction is lower K for REAL than NULL, or $\Delta K = K_N - K_R > 0$. In v25.6 this did **not** become a stable independent signal. It should therefore remain a control diagnostic.

chi - spectral susceptibility: how easily nearby spectral structure can mix under perturbation. The favourable direction is lower chi for REAL than NULL, or $\Delta \chi = \chi_N - \chi_R > 0$. This became the **strongest and most recurrent physical signal**.

Projector overlap (Ov) - measures how much the perturbed 2D subspace still coincides with the original 2D subspace. The favourable direction is $\Delta Ov = Ov_R - Ov_N > 0$. It is clearly useful, but more locally variable than chi.

Retention - asks whether the energy-matched perturbed neighbour pair still satisfies the gap criterion. In v25.5 it saturated at 1.0 for both REAL and NULL and therefore did not discriminate.

4. What v25.6 actually confirmed

v25.6 was deliberately not used to fit another continuation score. Its purpose was confirmation on independent seed blocks. The predeclared rule called a candidate STRONG when both the positive overlap rate and positive chi rate were at least 0.80; PARTIAL meant that chi passed while overlap did not.

Region 1: 153, 155, 157, 159

STRONG: 155, 157 and 159. PARTIAL: 153. The chi-positive rate was 1.000 at all four coordinates. Overlap-positive rates were 0.70, 0.90, 0.80 and 1.00. Thus 157 was not an isolated lucky point.

Region 2: 161, 163, 165, 167

Again chi-positive rate = 1.000 for all four candidates. Overlap was more uneven: 161 and 165 were STRONG (0.90), while 163 and 167 were PARTIAL (0.60 and 0.70). This is consistent with a broad physical region containing local internal structure rather than one smooth peak.

Region 4: 177, 179, 181, 183

Region 4 used base_seed 4401000 and therefore supplied an additional independent seed block. All four candidates were STRONG. Their overlap-positive rates were 1.00, 0.90, 0.80 and 0.80, and their chi-positive rates were all 1.000. Coordinate 177 was especially clean: even its minimum DeltaOv remained positive (+0.0004).

Record-keeping note: Region 3 formed part of the local scan sequence, but its detailed numerical output is not reproduced in the material used to prepare this note. No Region-3 numbers are invented here.

5. The central physical interpretation

The evidence no longer supports the simple picture '157 is the hotspot'. A better working picture is that REAL contains a broader region in which selected 2D subspaces are systematically less spectrally susceptible than their NULL counterparts, while the actual geometric retention of the subspace varies locally.

This suggests two layers of structure: (1) a broad physical background identified most consistently by chi, and (2) local robust cores identified by favourable projector overlap. These need not peak at exactly the same coordinate.

6. The relationship between K_delta, chi and overlap

The three quantities are not interchangeable. K_delta asks whether the perturbation directly couples the selected subspace to its complement. chi asks how susceptible the spectral environment is to mixing. Projector overlap asks what actually happens geometrically to the chosen 2D subspace after perturbation.

K_delta remained extremely close for REAL and NULL and did not reproduce with a stable favourable sign. Therefore it should not be promoted into the main Soft-Space criterion merely because it has an attractive theoretical interpretation.

chi behaved very differently. Its sign was highly recurrent: REAL repeatedly showed lower spectral susceptibility than NULL. Projector overlap then supplied a second, geometrical test of whether that reduced susceptibility was accompanied by better retention of the subspace. Overlap was positive often enough to be informative, but it was more coordinate- and seed-sensitive than chi.

Working hierarchy: chi identifies the broad protection tendency; overlap tests actual subspace retention; K_delta remains a diagnostic/control until independent evidence says otherwise.

7. Why the local scan should stop after Region 4

Continuing with 185, 187, 189 and so on could map the boundary more finely, but it would mainly add detail to a fact already established: the effect is not confined to coordinate 157. The scientific return from another neighbouring block is therefore lower than the return from testing whether the discovered physical signature survives a more demanding falsification protocol.

Stopping here also protects the analysis from post-hoc selection. Repeatedly moving the search window after seeing each result can gradually make the experiment depend on what has already looked favourable. The next experiment should instead freeze the hypotheses and test them on unseen data and appropriate controls.

8. What v25.7 should do

v25.7 should **not** be another local hotspot scan and should **not** immediately fit chi and overlap into a new weighted formula. Its first purpose should be **falsification and evidence strengthening**.

Design principle	Scientific reason
Predeclare Soft-Space candidates/region and control regions before execution.	Avoid choosing favourable locations after seeing results.
Use genuinely unseen seed blocks.	Test reproducibility, not reuse of the same random realization.
Keep the already defined signs of Deltachi and DeltaOv.	Freeze the hypothesis before the test.
Include fair geometry-matched controls away from the selected region.	Test whether the effect is specific rather than ubiquitous.
Use chi sign recurrence as the primary physical signal; overlap as independent geometric corroboration.	Reflect what v25.6 actually established.
Keep K_{Δ} as a control unless it begins to reproduce.	Do not add unsupported terms to the formalism.
Only after successful confirmation consider extending the continuation formula.	Separate discovery/validation from model fitting.

9. What would count as stronger evidence?

The strongest next result would not be one spectacular value at one coordinate. It would be a reproducible pattern on unseen seeds in which the preselected Soft-Space region shows positive Deltachi substantially more consistently than fair control regions, with projector overlap providing independent support. Conversely, if the same behaviour appears equally strongly in the controls, the present interpretation must be weakened or revised.

That is why v25.7 is scientifically more valuable than simply extending Region 4: it can potentially distinguish between a genuine REAL-specific physical structure and a generic property of the construction or diagnostic.

10. Relation to a future mathematical formalism

The present continuation expression should be regarded as a developing model, not a finished law. v25.4 showed that static empirical terms were insufficient. v25.5-v25.6 supplied candidate physical ingredients. The correct procedure is stepwise: discover a physically meaningful diagnostic, independently validate it, test it against controls, and only then decide whether and how it belongs in the mathematical continuation criterion.

This creates a natural bridge to open-system theory. Quantities such as susceptibility, subspace retention and coupling to the complement can later be related more formally to Hamiltonian perturbation theory and, where relevant, to the open-system framework associated with Breuer and Petruccione. That theoretical connection remains a programme for analysis; v25.6 by itself does not prove a Lindblad-based explanation.

11. Compact research conclusion

Current working conclusion: The 9-qubit experiments have moved the Soft-Spaces project beyond a purely empirical continuation of the old 8-qubit anchor. The strongest new observation is a recurrent REAL-specific reduction in spectral susceptibility (positive $\Delta\chi$), accompanied in many locations by improved projector overlap. K_{Δ} does not presently discriminate REAL from NULL and should remain a control quantity. The data are more consistent with a broad physical region containing locally stronger subspaces than with one isolated coordinate at 157. The proper next step is therefore v25.7: a predeclared, independent falsification/control experiment designed to test specificity and reproducibility before these physical quantities are promoted into an extended Soft-Space formula.

Research status note prepared from the recorded v25.4.1, v25.5 and v25.6 outputs available in the project conversation. Measured results, interpretation and the proposed v25.7 design are deliberately kept distinct.